

Math 344, F18
Exam 1

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	25	
3	20	
4	20	
5	15	
Total	100	

Good Luck !

Problem 1. (20 points) Given two vectors in the three dimensional space $\mathbf{v} = (3, \sqrt{3}, 2\sqrt{6})$ and $\mathbf{w} = (0, -\frac{2\sqrt{3}}{3}, \frac{2\sqrt{6}}{3})$.

(a) Find the length of $\mathbf{v}$ and $\mathbf{w}$.

$$\|\mathbf{v}\| = \sqrt{3^2 + (\sqrt{3})^2 + (2\sqrt{6})^2} = \sqrt{9 + 3 + 24} = \sqrt{36} = 6$$

$$\|\mathbf{w}\| = \sqrt{0^2 + (-\frac{2\sqrt{3}}{3})^2 + (\frac{2\sqrt{6}}{3})^2} = \sqrt{0 + \frac{4}{3} + \frac{8}{3}} = \sqrt{4} = 2$$

(b) Find the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

$$\mathbf{v} \cdot \mathbf{w} = 3 \cdot 0 + \sqrt{3} \cdot (-\frac{2\sqrt{3}}{3}) + 2\sqrt{6} \cdot \frac{2\sqrt{6}}{3} = 0 - 2 + 8 = 6$$

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \cdot \|\mathbf{w}\|} = \frac{6}{6 \cdot 2} = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3} \text{ or } 60^\circ$$

(c) Find a unit vector $\mathbf{u}$ which has the same direction as $\mathbf{v}$.

$$\begin{aligned} \mathbf{u} &= \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{(3, \sqrt{3}, 2\sqrt{6})}{6} \\ &= (\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3}) \end{aligned}$$

(d) Find a number α so that $\mathbf{w} + \alpha\mathbf{v}$ is perpendicular to $\mathbf{v}$.

$$\begin{aligned} (\mathbf{w} + \alpha\mathbf{v}) \cdot \mathbf{v} &= 0 \\ \mathbf{w} \cdot \mathbf{v} + \alpha \|\mathbf{v}\|^2 &= 0 \\ \Rightarrow \alpha &= -\frac{\mathbf{w} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \\ &= -\frac{6}{6^2} \\ &= -\frac{1}{6} \end{aligned}$$

Problem 2. (25 points)

(a) Find an 2×2 matrix $\mathbf{A}$ such that $\mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2x - 3y \end{bmatrix}$.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix}$$

(b) Find an 3×3 elimination matrix $\mathbf{E}$ that adds 4 times of row 1 to row 3.

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 0 \end{bmatrix}$$

(c) Find an 3×3 permutation matrix $\mathbf{P}$ that exchanges rows 1 and 3, then rows 3 and 2.

$$\begin{aligned} \mathbf{P} &= \mathbf{P}_{32} \mathbf{P}_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{aligned}$$

(d) Find the inverse of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ if it is invertible.

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{-1} &= \frac{1}{1 \cdot 4 - 2 \cdot 3} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \frac{1}{(-2)} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \end{aligned}$$

(e) Find two 2×2 matrices, $\mathbf{A}$ and $\mathbf{B}$, such that $\mathbf{A} \neq \mathbf{0}$ and $\mathbf{B} \neq \mathbf{0}$ but $\mathbf{AB} = \mathbf{0}$.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Problem 3. (20 points) Given $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ -2 & -2 & 0 \end{bmatrix}$, find A^{-1} using the Gaussian-Jordan

Elimination.

$$[A \ I] = \begin{bmatrix} \textcircled{1} & 1 & 2 & | & 1 & 0 & 0 \\ \textcircled{1} & 3 & 4 & | & 0 & 1 & 0 \\ -2 & -2 & 0 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} \textcircled{1} & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 2 & 2 & | & -1 & 1 & 0 \\ -2 & -2 & 0 & | & 0 & 0 & 1 \end{bmatrix} \left. \vphantom{\begin{bmatrix} \textcircled{1} & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 2 & 2 & | & -1 & 1 & 0 \\ -2 & -2 & 0 & | & 0 & 0 & 1 \end{bmatrix}} \right\} \text{Gaussian}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 2 & \textcircled{2} & | & -1 & 1 & 0 \\ 0 & 0 & \textcircled{4} & | & 2 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & \textcircled{2} & | & 1 & 0 & 0 \\ 0 & 2 & 0 & | & -2 & 1 & -\frac{1}{2} \\ 0 & 0 & \textcircled{4} & | & 2 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \textcircled{1} & 0 & | & 0 & 0 & -\frac{1}{2} \\ 0 & \textcircled{2} & 0 & | & -2 & 1 & -\frac{1}{2} \\ 0 & 0 & 4 & | & 2 & 0 & 1 \end{bmatrix} \left. \vphantom{\begin{bmatrix} 1 & \textcircled{1} & 0 & | & 0 & 0 & -\frac{1}{2} \\ 0 & \textcircled{2} & 0 & | & -2 & 1 & -\frac{1}{2} \\ 0 & 0 & 4 & | & 2 & 0 & 1 \end{bmatrix}} \right\} \text{Jordan}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & | & 1 & -\frac{1}{2} & -\frac{1}{4} \\ 0 & 2 & 0 & | & -2 & 1 & -\frac{1}{2} \\ 0 & 0 & 4 & | & 2 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & | & 1 & -\frac{1}{2} & -\frac{1}{4} \\ 0 & 1 & 0 & | & -1 & \frac{1}{2} & -\frac{1}{4} \\ 0 & 0 & 1 & | & \frac{1}{2} & 0 & \frac{1}{4} \end{bmatrix} \leftarrow \text{scaling}$$

$$= [I \mid A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{4} \\ -1 & \frac{1}{2} & -\frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{4} \end{bmatrix}$$

Problem 4. (20 points) Let $A = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix}$ and $b = \begin{bmatrix} -6 \\ 4 \\ -2 \end{bmatrix}$.

(a) Find the LU and LDU factorizations of A .

$$\begin{aligned}
 A &= \begin{bmatrix} \boxed{-1} & 1 & -3 \\ \boxed{2} & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{l_{21}=-2} \begin{bmatrix} \boxed{-1} & 1 & -3 \\ 0 & -2 & -5 \\ \boxed{-3} & -3 & -4 \end{bmatrix} \xrightarrow{l_{31}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & \boxed{-2} & -5 \\ 0 & \boxed{-6} & 5 \end{bmatrix} \\
 &\xrightarrow{l_{32}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix} \\
 \Rightarrow A &= LU = \begin{bmatrix} 1 & & \\ -2 & 1 & \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & -3 \\ & -2 & -5 \\ & & 20 \end{bmatrix} \\
 &= LDU = \begin{bmatrix} 1 & & \\ -2 & 1 & \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & & \\ & -2 & \\ & & 20 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \\ & 1 & \frac{5}{2} \\ & & 1 \end{bmatrix}
 \end{aligned}$$

(b) Using the above LU factorization to solve $Ax = b$ by solution of two triangular systems $Lc = b$ and then $Ux = c$.

$$\begin{aligned}
 Lc &= b \Rightarrow \begin{bmatrix} 1 & & \\ -2 & 1 & \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -6 \\ 4 \\ -2 \end{bmatrix} \Rightarrow \begin{cases} c_1 = -6 \\ -2c_1 + c_2 = 4 \\ 3c_1 + 3c_2 + c_3 = -2 \end{cases} \\
 \Rightarrow c_1 &= -6 \\
 c_2 &= 4 + 2c_1 = -8 \\
 c_3 &= -2 - 3c_1 - 3c_2 = 40 \\
 Ux &= c \Rightarrow \begin{bmatrix} -1 & 1 & -3 \\ & -2 & -5 \\ & & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -6 \\ -8 \\ 40 \end{bmatrix} \Rightarrow \begin{cases} -x_1 + x_2 - 3x_3 = -6 \\ -2x_2 - 5x_3 = -8 \\ 20x_3 = 40 \end{cases} \\
 \Rightarrow x_3 &= 2 \\
 -2x_2 &= -8 + 5x_3 = 2 \Rightarrow x_2 = -1 \\
 -x_1 &= -6 - x_2 + 3x_3 = 1 \Rightarrow x_1 = -1 \\
 \Rightarrow x &= \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}
 \end{aligned}$$

Problem 5. (15 points) True or False (no explanation needed).

(a) If a matrix $\mathbf{A}$ is invertible, then $\mathbf{A}^2$ must be invertible.

True $(\mathbf{A}^2)^{-1} = (\mathbf{A} \cdot \mathbf{A})^{-1} = \mathbf{A}^{-1} \mathbf{A}^{-1}$

(b) If $\mathbf{A}$ and $\mathbf{B}$ are symmetric, then their product $\mathbf{AB}$ must be symmetric.

False $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
 $\mathbf{AB} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$ not symmetric

(c) Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, there exists a 3×3 matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{I}$.

False $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ no second pivots
 $\rightarrow$ not invertible

(d) The column space of $2\mathbf{A}$ equals the column space of $\mathbf{A}$ where $\mathbf{A}$ is a matrix.

True $\mathbf{AX} = 2\mathbf{A}\left(\frac{\mathbf{X}}{2}\right)$

(e) If a subspace of the vector space $\mathbf{M}$ contains $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$, then it must contain the identity matrix $\mathbf{I}$.

True $\frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$